

A Canonical Correlation Analysis of Global Economic Production and Expenditure

STAT 348 - Professor Severini

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1 Explanation of the Data

This multivariate statistical analysis is based on a set of economic indicators for countries across the globe during 2019. The dataset was taken from a Kaggle website entitled “Global Economy Indicators”, which compiled data from the Economic Statistics Branch of the United Nations Statistics Division, as well as various other international statistics organizations (“Global Economy”). To provide an overview of the contents of the data, the observations and variables contained in the data will be explained.

Each observation represents a single country in the year 2019. There are a total of $n = 208$ countries and territories represented in the data that were used during the analysis. 212 countries and territories are present in the original dataset, however four were removed for having missing data: North Korea, which had many missing values, likely due to a lack of public reporting, as well as Macao, Monaco, and Sint Maarten, all of which are small enough countries (in terms of both population and area) to disregard in the analysis. The 208 used in the analysis include 191 out of the 193 UN member states (North Korea and Monaco are removed) as well as other territories, and therefore represents a very large portion of the globe.

Each variable represents a different economic indicator associated with a country at the end of 2019. There are a total of $p = 12$ variables in the dataset, each constituting a different measure of a country’s economy. It is important to note that the 12 variables can be partitioned into two sets of variables, both uniquely representing a country’s total GDP. Each variable is given below, along with an explanation. All quantities are based on 2019 US dollars.

1. Household Consumption Expenditure (as a % of GDP) – total spending by all of a country’s households divided by its GDP. This represents how much a country’s citizens are consuming relative to its total GDP.
2. Gross Capital Formation (GCF) (as a % of GDP) – the total investment in a country divided by its GDP. GCF (also known as investment) is a measure of how much a country spends on capital that will help increase production in the future, and is not

directly consumed. This includes machinery, factories, buildings, rollover inventories, etc.

3. General Government Final Consumption Expenditure (as a % of GDP) – total spending by a country’s government divided by its GDP. This quantity represents how involved a government is in their country’s economy.
4. Exports (as a % of GDP) – all exports divided by the GDP of a country. This represents the dependency of a country’s economy on its exports, or goods and services that are sold to a foreign country.
5. Imports (as a % of GDP) – all imports divided by the GDP of a country. This represents the dependency of a country’s economy on its imports, or goods and services that are bought from a foreign country.

The above 5 variables collectively represent the five different components of GDP according to the “Expenditure Method” of calculating GDP, whose formula is given by:

$$\text{GDP} = \text{Household Consumption} + \text{Investment} + \text{Government Expenditure} + \text{Net Exports}$$

where $\text{Net Exports} = \text{Exports} - \text{Imports}$ (Fernando).

Since the dataset used for this analysis includes these five variables as percentages of GDP, a linear combination of the variables according to the above equation should be roughly equal to 100 for all countries in the dataset. Together, these 5 variables answer the question: how is a country consuming its GDP?

6. Agriculture, hunting, forestry, and fishing output (as a % of GDP) – the contribution of all agriculture-, hunting-, forestry-, and fishing- related activities, as defined by ISIC, to the GDP of a country.
7. Construction output (as a % of GDP) – the contribution of all construction activities, as defined by ISIC, to a country’s GDP.
8. Mining, manufacturing, and utilities output (as a % of GDP) – the contribution of the mining, manufacturing, and utilities industries, as defined by ISIC, to a country’s GDP.
9. Manufacturing output (as a % of GDP) – the contribution of the manufacturing industry, as defined by ISIC, (industrialized, manufacturing activities) to a country’s GDP.
10. Transport, storage, and communication output (as a % of GDP) – the contribution of transportation, storage, and communication industries, as defined by ISIC, to a country’s GDP.
11. Wholesale, retail, trade, restaurants, and hotels output (as a % of GDP) – the contribution of wholesale, retail, trade, restaurants, and hotels, as defined by ISIC, to a country’s GDP.
12. Other activities/services output (as a % of GDP) – the contribution of all other economic activity as defined by ISIC (including mainly services industries such as finance, healthcare, etc.).

Similarly to how the first 5 variables combine to form GDP, a linear combination of the above 7 variables can form a country’s total GDP, though it is being calculated from a different perspective. Specifically, the sum of these variables for a country should be equal to 100, as they collectively form all elements of GDP from the perspective of different industries

producing output. This is referred to as the “Production Method” of calculating GDP (Statistics Canada), and is defined by the formula:

$$GDP = \sum_i \text{total value added for sector } i$$

Collectively, these 7 variables answer the question: how is a country producing its GDP?

It may seem as though there is some overlap between the 7 variables presented above - for example, between the categories “Manufacturing output” and “Mining, manufacturing, and utilities output”, as they both seem to contain economic output as a result of manufacturing. However, these variables are based on the International Standard Industrial Classification of All Economic Output (ISIC) as crafted by the United Nations, and are mostly mutually exclusive with minimal overlap (UN Statistics). Although there are 21 total ISIC sectors, some have been aggregated here (such as the “Other activities/services output” variable) to represent the total economic output of each country. The most important consideration is that these variables are consistent for all countries, which allows for a robust, interpretable multivariate analysis.

Thus, the dataset consists of two different sets of variables that are each a unique representation of the composition of a country’s GDP. The “expenditure” variables explain how different market participants spend GDP (consumers, governments, businesses, trade), whereas the “production” variables explain which sectors produce GDP (construction, agriculture, manufacturing, etc.). An important note about the data is that although the values in both sets of data theoretically could linearly combine to 100, they do not for most countries due to imperfections in the data.

One final important consideration for the dataset is why 2019 was chosen as the year whose data would be analyzed. 2019 was selected for two main reasons. Firstly, 2019 has very few missing (or NA) values as compared to other years; since all rows with missing values were dropped, choosing a year with as few missing values as possible became extremely important for the analysis to consider many countries around the world. Secondly, 2019 represents a period of typical, stable global economic growth without any major disruptions. While both the early 2010s and the 2020s are characterized by recovery from global economic catastrophes (the recession caused by the 2008 housing crash and the coronavirus pandemic, respectively), 2019 is the most recent year in which the economy exhibited its typical patterns, undisrupted by major economic downturns or unusually high growth.

2 Goals of the Analysis

The primary goal of this analysis is to investigate the question: “Are there any strong or moderately strong relationships between the ways that a country produces its GDP and the ways that it spends its GDP? Are these relationships significant? What can these relationships tell us about the global economy?”.

Basic macroeconomic intuition suggests that there will likely be associations between a country’s methods of production and their expenditures. For example, if a country’s primary

way of producing GDP is through manufacturing, it might be more inclined to export the goods they manufacture to another country in order to profit. Countries with small agricultural industries and large services industries may have high imports, as they may need to purchase their food from abroad. The goal of this analysis is precisely to identify whether these types of strong relationships - between specific methods of production and specific types of expenditure - exist in the data, and evaluate their strength and significance.

3 Summary of the Analysis

A natural first step in analyzing the relationship between these two sets of variables is to calculate the cross-correlation matrix, which contains all pairwise correlations between the variables in each set. Since the number of variables in each of the two sets of data is small (5 and 7), examining all of the pairwise correlations is simple to do. A table containing the rounded pairwise cross-correlations is printed below.

	Agri	Constr	Manuf	MinManUtil	TransStorComm	WholRetTra	Serv
Cons_house	0.48	-0.14	-0.15	-0.34	-0.04	0.18	-0.24
Cons_gov	-0.08	0.06	-0.21	-0.14	0.05	-0.13	0.30
Invest	0.04	0.31	0.05	0.07	0.08	-0.08	-0.20
Export	-0.42	-0.13	0.10	0.03	0.23	0.05	0.29
Import	-0.20	-0.11	-0.05	-0.19	0.27	0.08	0.23

There are no strong pairwise correlations between the two datasets. The largest-magnitude correlation coefficient is 0.48 between Agriculture and Household Consumption, indicating a moderate positive linear relationship between the two variables. The second largest-magnitude correlation is -0.42 between Agriculture and Exports, exhibiting a moderate negative relationship. All other pairwise correlation coefficients are smaller than 0.35 in magnitude, indicating weak linear relationships, and thus are not worth exploring.

An extension of analyzing pairwise correlations between two individual variables is to instead analyze correlations between linear combinations of the variables in each set of data. This is done through canonical correlation analysis. In the context of the data, we can let the random vector X denote the full set of 12 economic variables with $X^{(1)}$ being a partition of X containing only the 5 expenditure variables (centered around their means) and $X^{(2)}$ being a partition of X containing only the 7 production variables (centered around their means). Canonical correlation analysis finds the two linear combinations, $U_1 = a'X^{(1)}$ and $V_1 = b'X^{(2)}$ called the first canonical variables, which maximize $\text{Corr}(U_1, V_1)$, called the first canonical correlation. It then finds another pair of canonical variables U_2 and V_2 (of similar form to U_1 and V_1) that again maximizes the canonical correlation $\text{Corr}(U_2, V_2)$ subject to the restriction that they are uncorrelated with U_1 and V_1 . The method continues to find a total of $\min(\dim(X^{(1)}), \dim(X^{(2)})) = \min(5, 7) = 5$ pairs of canonical correlations and their corresponding pairs of canonical variables, each pair being uncorrelated with all previous ones. Examining the coefficients of one pair of canonical variables would yield an interpretation of the relationship between the production and expenditure variables. The magnitude of the corresponding canonical correlation provides a measure of the strength of that particular relationship.

After conducting canonical correlation analysis through the procedure described above, the first two canonical correlations are 0.65 and 0.50. All of the following canonical correlations are less than 0.40, and thus do not represent strong relationships between the production and expenditure variables. Hence, only the first two pairs of canonical variables will be explored in depth. The full R output from the canonical correlation analysis, including all 5 pairs of canonical variable coefficients and their corresponding canonical correlations, are located in the “Supporting Materials & Numerical Outputs” section at the end of this report.

The first pair of canonical variables has a canonical correlation of 0.65. Since only the relative magnitudes and signs of the coefficients for the canonical variables matter, a rescaling of both canonical variable’s coefficients on the original expenditure variables can be obtained and lead to the same canonical correlation. A rescaled approximated version of the first expenditure canonical variable is:

$$U_1 = \text{Imports} - 5 * \text{Household Consumption} - 5 * \text{Investment} - \text{Exports}$$

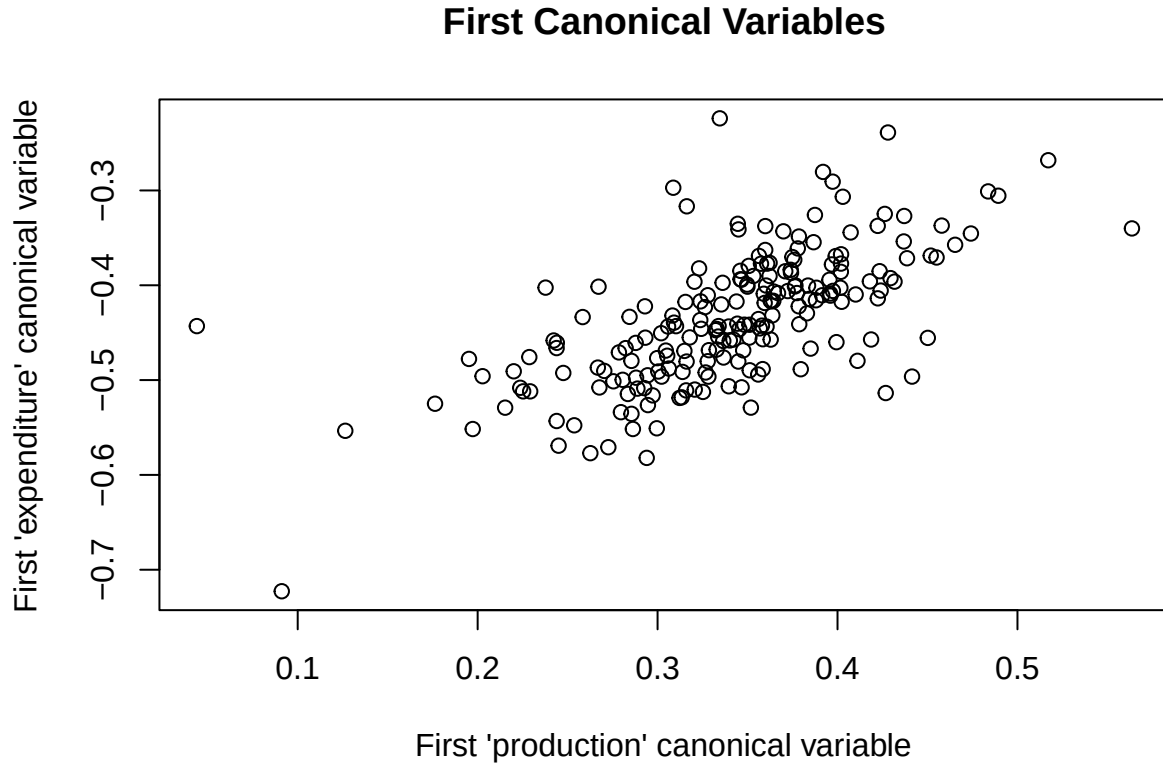
Imports is the only expenditure variable with a positive coefficient, while Household Consumption, Investment, and Exports all have negative coefficients. Household Consumptions and Investment have particularly large negative coefficients. Government Expenditure was excluded from this approximated rescaling of the first expenditure canonical variable because its coefficient was extremely small in comparison to the other coefficients. Hence, this variable increases when a country has a high share of imports in its GDP and a low share of household consumption, investment, and exports in its GDP, while government expenditure barely affects the canonical variable. This variable could be interpreted as contrasting a country’s dependency on trade with its own domestic economic activity, with particular emphasis on household consumption and investment.

Similarly, a rescaled approximated version of the first production canonical variable is:

$$V_1 = 2 * \text{Construction} + 5 * \text{Mining \& Utilities} + 4 * \text{Transport \& Comms.} \\ + 2 * \text{Retail \& Restaurants} + 5 * \text{Services} - \text{Manufacturing} - \text{Agriculture}$$

Manufacturing and Agriculture are the only variables with negative coefficients, while all other variables have positive coefficients. This variable thus increases when Manufacturing and Agriculture represent small shares of a country’s GDP relative to the rest of the country’s industries. Particularly, countries with large Utilities, Transportation, and Services industries (all the high-coefficient positive variables) will have large values of this variable. Hence, this variable can be seen as a measure of the country’s post-industrial development, with large values representing more service-, utilities- and transportation-driven economies and small values representing more industrial-driven economies.

The first canonical correlation of 0.65 indicates that the two variables described above exhibit moderate to strong positive correlation - as one variable increases, the other generally increases as well. A scatter plot containing the observed values of the first production canonical variable against the first expenditure canonical variable is printed below.



As expected, there is a fairly strong upward trend. Taking into account the previous interpretations of the first pair of canonical variables, this scatter plot shows that as a country's economy shifts away from relying on industrial sectors like manufacturing and agriculture and towards post-industrial sectors like services, transportation, and utilities, the share of imports in their GDP tends to increase, and domestic economic activity such as household consumption, investment, and exports tends to decrease. Given the first canonical correlation of 0.65, the data indicates a moderate to strong relationship in this trend.

A similar process can be followed to interpret the second pair of canonical variables, which have a canonical correlation of 0.50. A rescaled approximation of the second expenditure canonical variable is given by:

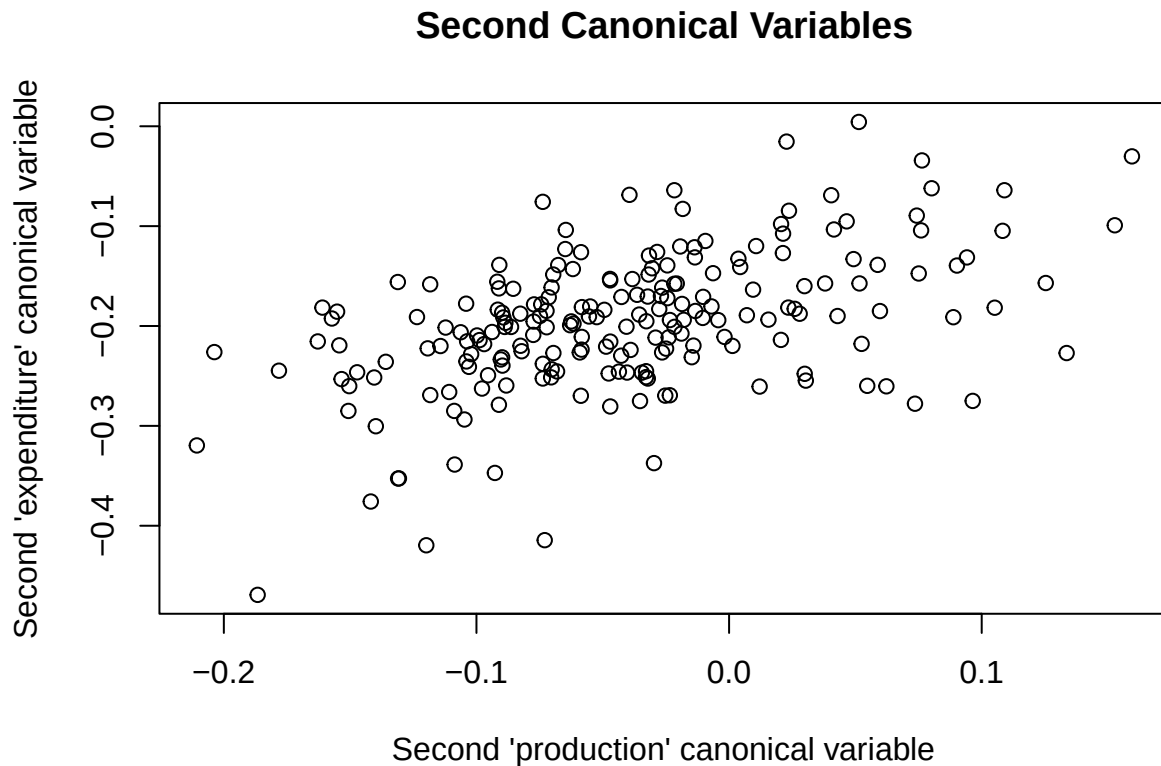
$$U_2 = 2 * \text{Investment} - 2 * \text{Household Consumption} - \text{Gov. Consumption} - \text{Exports}$$

This variable is large when a country has a high share of investment in its GDP, and lower spending in the other GDP components, particularly in household consumption. There is virtually zero weight given to the share of imports in GDP. Thus, this variable can be interpreted as a contrast between a country's investment (into future output) and its current spending - it takes on large values if a country is focused on investing capital into their future growth as opposed to spending their money now.

A rescaled approximation of the second production canonical variable is:

$$V_2 = 3 * \text{Construction} - \text{Transport \& Comms.}$$

This variable increases when the share of construction in GDP increases, and decreases with transportation and communication shares in GDP. Given the extremely small coefficients of all other production variables, this canonical variable can be interpreted to represent how heavily a country's GDP is dependent on construction as opposed to other sectors (specifically transportation and communication). A scatter plot of the observed values of the second production canonical variable against the second expenditure canonical variable is seen below.



The scatterplot, along with the second canonical correlation of 0.50, indicates that the relationship between the second pair of canonical variables is moderately supported by the data, but it is not nearly as strong as the relationship presented by the first pair of canonical variables. Using the interpretations of the second pair of canonical variables, this means that with moderate strength, as the size of the construction sector increases relative to other sectors within a country, especially transportation and communication, the country tends to invest more money into future output as opposed to spending it through household and government consumption.

After interpreting the first two canonical correlations and determining that the relationships between the production and expenditure variables presented in the canonical variables have at least a moderate degree of strength, it is natural to ask whether the correlations themselves are significant given the amount of data presented. This question can be answered through a series of sequential tests of the canonical correlations, which asks, for all 5 canonical correlations, whether all remaining canonical correlations are collectively equal to 0, or whether at least one remaining canonical correlation is non-zero. A more complete explanation of this method

is given in the “Appendix” section of this report.

Conducting the sequential tests for significance of the canonical correlations results in the following p-values for each of the canonical correlations, respectively: 0, 2.81e-12, 2.42e-05, 1.30e-02, and 4.63e-01. At an $\alpha = 0.05$ level of significance, the first four canonical correlations are significantly different from 0, and the fifth is not. Thus, it would be appropriate to interpret the first four pairs of canonical variables. However, this sequential test procedure is oftentimes not suggested to be used as a definitive rule for determining which canonical correlations are significant, and should instead only be taken as a rough guideline for determining which correlations may be important to analyze. Taking this into consideration, alongside the goal of this analysis - which is to identify relationships between the expenditure and production variables that are both significant and strong - only the first two canonical correlations will be interpreted, as they are the only two that are both significant and exhibit at least a moderate degree of strength.

4 Conclusions

The canonical correlation analysis reveals that there are exactly two meaningful and statistically significant relationships between a country’s methods of producing GDP and their expenditure of GDP. These two relationships, captured by the first two pairs of canonical variables, are the only ones that demonstrate at least moderate strength, indicating that much of the systematic connection between expenditure and production can be understood through these two dimensions.

The first relationship shows that countries with higher proportions of their GDP being produced through the services, utilities, and transportation sectors typically have a higher level of imports and less domestic economic activity, such as household consumption, investment, and exports. This dynamic highlights a global distinction between economies that are more trade-dependent and services-focused, and those that remain largely industrially- or agriculturally-driven. Hence, this relationship reflects two very broad stages of economic development: as less-developed countries with agrarian and industrial structures driving their output shift into more services-oriented economies, they become more dependent on purchasing items from abroad and less dependent domestic activities like consumption and investment.

The second relationship focuses on countries where the construction industry comprises a large portion of GDP. The data supports that such countries oftentimes allocate more of their GDP to investment in future output rather than spending it through consumption, government expenditures, or trade. This pattern reflects economic intuition: it shows that investments in future output often involve the construction of physical assets such as buildings, factories, and other substantial construction projects. Therefore, this dimension captures economies in phases of physical expansion, urbanization, and infrastructure-driven growth. It is also important to note that although this pattern is evident in the data, it is less strongly-supported than the first relationship.

When taken together, these two relationships address the core questions that motivated this

analysis. There are two identifiable and economically meaningful patterns evident in the data that connect the ways in which a country produces its GDP to the ways it spends it. Importantly, the statistical significance and moderate strength of these relationships show that these findings are robust and rooted in real structural differences across national economies.

5 Appendix

The sequential test for significance of canonical correlations, which was used in the “Summary of the Analysis” section of this report, is explained below, and was written while referencing section 10.6 of the textbook *Applied Multivariate Statistical Analysis* by Johnson and Wichern (Johnson).

The test procedure assumes that the random vector X follows a multivariate normal distribution with mean μ and covariance matrix Σ . The random vector, mean vector, and covariance matrix can all be partitioned such that we obtain a cross-covariance matrix, Σ_{12} of dimension p, q . A typical canonical correlation analysis can be conducted on the random vector, resulting in $\min(p, q)$ canonical correlations. In the following, we assume for the sake of simpler notation that $p < q$, and thus there are p canonical correlations.

The first step is to test whether $\Sigma_{12} = 0$, which would imply that all canonical correlations are also 0. The hypotheses are thus:

$$H_0 : \rho_1^* = \rho_2^* = \dots \rho_p^* = 0$$

$$H_1 : \text{at least one } \rho_i^* \neq 0 \text{ for } i = 1, \dots, p$$

The test statistic is based on the likelihood ratio test statistic of $-n \ln \prod_{i=1}^p (1 - \hat{\rho}_i^{*2})$, however it is slightly modified to better approximate the $\chi_{p,q}^2$ distribution. The portion of the test statistic containing the canonical correlations is $-\ln \prod_{i=1}^p (1 - \hat{\rho}_i^{*2})$, which is small when all the canonical correlations are small and is large when all the canonical correlations are large. Thus, the null hypothesis will be rejected when the canonical correlations are large, and will fail to be rejected when the canonical correlations are small.

The exact decision rule is to reject the null hypothesis at the α level of significance when

$$-(n - 1 - \frac{1}{2}(p + q + 1)) \ln \prod_{i=1}^p (1 - \hat{\rho}_i^{*2}) > \chi_{p,q}^2(\alpha)$$

or when the corresponding p-value is smaller than α . Given that this first test is rejected (which it often is for most data), we can continue to test each of the canonical correlations in order of largest to smallest magnitude. If the first test was rejected, we know that the “at least one” canonical correlation that must be non-zero would be the largest one. Thus, we continue by testing each of the p canonical correlations largest to smallest through a sequence of tests given by the hypotheses:

$$H_0^k : \rho_1^* \neq 0, \rho_2^* \neq 0, \dots, \rho_k^* \neq 0, \rho_{k+1}^* = \dots = \rho_p^* = 0$$

H_1^k : at least one $\rho_i^* \neq 0$ for $i \geq k + 1$

The null hypothesis for each test is rejected if

$$-(n - 1 - \frac{1}{2}(p + q + 1)) \ln \prod_{i=k+1}^p (1 - \hat{\rho}_i^{*2}) > \chi_{p-k, q-k}^2(\alpha)$$

We continue to test each H_0^k until one fails to be rejected at the given significance level.

This test relies on a set of assumptions that make it difficult to interpret when violated (especially multivariate normality). It is oftentimes not advisable to select a given level of significance and deem all canonical correlations after which the test fails to reject the null hypothesis as “insignificant”. Instead, this test should only be used only as a guide for understanding how many canonical correlations may be useful in an analysis.

In R, the “p.asym” function from the CCP package is used to calculate the p-values of these tests.

6 References

Fernando, Jason. “Gross Domestic Product (GDP) Formula and How to Use It.” Investopedia, 18 Oct. 2025, www.investopedia.com/terms/g/gdp.asp.

“Global Economy Indicators.” Kaggle, 9 Oct. 2023, www.kaggle.com/datasets/prasad22/global-economy-indicators.

Johnson, Richard, and Dean Wichern. Applied Multivariate Statistical Analysis. Pearson Prentice Hall, 2007.

Statistics Canada. Gross Domestic Product by Production Approach, 23 Jan. 2024, www150.statcan.gc.ca/n1/pub/13-607-x/2016001/175-eng.htm.

UN Statistics Division. “ISIC.” United Nations, unstats.un.org/unsd/classifications/Econ/isic.

7 Supporting Material & Numerical Outputs

```
### Cross-Correlation Matrix (unrounded)
cor(expenditure_vars, production_vars)
```

```
##           Agri      Constr      Manuf  MinManUtil  TransStorComm
## Cons_house 0.48184123 -0.13569939 -0.15338822 -0.34278431 -0.03859650
## Cons_gov   -0.08487634  0.05738962 -0.20639598 -0.13968141  0.04876219
## Invest     0.04118122  0.30889246  0.04856575  0.06657606  0.08277515
## Export     -0.41769641 -0.13415691  0.10179620  0.03383395  0.22774160
## Import     -0.19686428 -0.10627918 -0.04547220 -0.19237509  0.27155602
##           WholRetTra      Serv
## Cons_house 0.17670218 -0.2374340
```

```
## Cons_gov    -0.12781119  0.3035797
## Invest      -0.07915797 -0.1957358
## Export      0.04680843  0.2918620
## Import      0.08099514  0.2325742
```

Canonical Correlation Analysis Output

```
cancor(expenditure_vars, production_vars)
```

```
## $cor
## [1] 0.6536457 0.5007370 0.3656338 0.2835159 0.1127692
##
## $xcoef
##           [,1]      [,2]      [,3]      [,4]      [,5]
## Cons_house -5.208207e-03 -0.0027837157 -0.005792836 0.0003441707 -0.03410073
## Cons_gov    7.587038e-05 -0.0015141379 -0.011052733 0.0007460951 -0.03308692
## Invest     -5.377282e-03  0.0035240324 -0.005630601 0.0052155743 -0.03446754
## Export     -1.056056e-03 -0.0015274795 -0.004865099 0.0001231896 -0.03479347
## Import      1.441361e-03 -0.0002964283  0.005863727 0.0013914928  0.03464565
##
## $ycoef
##           [,1]      [,2]      [,3]      [,4]
## Agri       -0.000759705 -6.694027e-04 -0.0079809741 -0.0039842093
## Constr      0.001750398  1.474015e-02 -0.0111031549  0.0061145217
## Manuf      -0.001333354  5.991584e-05  0.0039075082  0.0008744628
## MinManUtil  0.005204103  2.446279e-03 -0.0040212483 -0.0046955721
## TransStorComm 0.004180347 -5.200792e-03 -0.0046403452  0.0137440699
## WholRetTra  0.001855837 -3.365355e-03 -0.0008759293 -0.0063925032
## Serv       0.005623133 -2.244118e-03 -0.0082654862 -0.0034890409
##
##           [,5]      [,6]      [,7]
## Agri       -0.0062951119 -0.0025857938  0.0104604374
## Constr     -0.0099163517 -0.0121679427  0.0049822935
## Manuf      -0.0107774000  0.0004043602 -0.0006435257
## MinManUtil -0.0008670848 -0.0028927224  0.0101032979
## TransStorComm -0.0023212644  0.0012814098  0.0249285060
## WholRetTra -0.0049212302 -0.0132871822  0.0107057286
## Serv      -0.0069824956 -0.0041294097  0.0082898779
##
## $xcenter
## Cons_house  Cons_gov    Invest    Export    Import
##   63.07001   18.12488   24.76610   43.24159   49.25734
##
## $ycenter
##           Agri           Constr           Manuf           MinManUtil TransStorComm
##   10.127971    5.912932   11.255517   19.349229    8.274764
##   WholRetTra           Serv
```

```
##      15.049612      33.738903
```

```
### Sequential tests of canonical correlations
```

```
library(CCP)
```

```
p.asym(  
  cancel(expenditure_vars, production_vars)$cor,  
  nrow(expenditure_vars),  
  ncol(expenditure_vars),  
  ncol(production_vars)  
)
```

```
## Wilks' Lambda, using F-approximation (Rao's F):
```

```
##           stat      approx df1      df2      p.value  
## 1 to 5:  0.3375366 6.9598170  35 826.9273 0.000000e+00  
## 2 to 5:  0.5893290 4.6947821  24 688.4609 2.805978e-12  
## 3 to 5:  0.7865455 3.3137306  15 546.9922 2.418227e-05  
## 4 to 5:  0.9079241 2.4617588   8 398.0000 1.301867e-02  
## 5 to 5:  0.9872831 0.8587123   3 200.0000 4.634697e-01
```